

Dakshi Goel



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EDUCATION

Bachelor of Technology, Indian Institute of Technology (Jammu, India)

July 2020 – May 2024

Majors - Electrical Engineering | Minors - Computer Science

GPA - 8.53/10.0

SKILLS

Programming Skills: Python, C++, SQL, NodeJS, ReactJS

Relevant Coursework: Database Management System, Machine Learning, Deep Learning, Data Structures and Algorithms, Digital Image Processing, Calculus, Linear Algebra, and Differential Equations.

Libraries/Platforms: NLTK, LangChain, Llama-Index, OpenAI, Gemini, Claude, FastAPI, REST API, BeautifulSoup, AWS, Postman, Pandas, Numpy, Matplotlib, Sklearn, TensorFlow, Keras, PyTorch, Vercel.

PUBLICATIONS

Illumination Aware Multi-scale Attention Fusion Model for Underwater Image Enhancement

Accepted in *IEEE Transactions on Artificial Intelligence* | Indian patent filed ([No.202411053490](#))

- Addressed class imbalance by combining SMOTE and GANs, achieving a 2-3% improvement in F1 scores for classification.
- Achieved accuracy of 98.4% (CNN MLP) for predicting alteration zones for remote sensing data.

Remote sensing data processing using convolutional neural networks for mapping alteration zones

2024 *International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing (MIGARS)*

- Created a multi-scale U-Net architecture outperforming SOTA methods by 1.18 dB in PSNR and 0.948 in SSIM.
- Performed quantitative analysis with 14 models over 8 evaluation metrics for image quality.

WORK EXPERIENCE

Real Brokerage, AI Product Associate (Remote)

Dec 2024 - Current

- Analyzed 10,000+ chatbot conversations through Langfuse to resolve intent routing failures, using pattern matching (regex) and vector search. Implemented structured Q-tags to enrich chunks. Increased retrieval and routing accuracy by 20%.
- Enhanced LLM generation by designing few-shot, table-structured prompts and refining API trigger logic with context-aware reranking, reducing null/incorrect responses by 25% while improving end-to-end chatbot reliability.

Mozrest, Junior AI Developer (Remote)

Dec 2023 - Dec 2024

- Designed and deployed a GPT-3.5 Turbo for a task-oriented dialogue system with LangGraph on WhatsApp, integrating Stripe, and built a review response generation tool using few-shot prompting with sentiment control using GPT.
- Built and deployed forecasting models (ARIMA + LSTM) for restaurant sales and slot availability, achieving RMSE 7.2 and 93% accuracy while reducing latency by 125 ms through optimized 3,000-feature pipelines on AWS Lambda.

ML Course Mentor, IIT Jammu

- Mentored 50+ students across batches, guided hands-on model implementation and code optimization during labs.
- Designed 6+ assignments, prepared lecture materials, and led 3+ research paper discussions to strengthen foundational.

LTI Mindtree, Project Intern (Bangalore, India)

May 2023 - July 2023

- Implemented RAG pipeline and fine-tuned Llama2 13B model for code generation for Snowpark.
- Reduced hallucination rates by 40% using chain-of-thought prompting. Developed UI using Streamlit and FastAPI.

PROJECTS

Voice Agent for Real Estate (Demo)

- Built a voice-powered real estate search assistant (Node.js, Vapi.ai), with hybrid semantic search across 100+ US properties.
- Containerized ranking system (Docker, Vercel) with real-time conversational updates and smart recommendations.

Flipkart Grid Challenge

- Developed text-to-SQL LLM to interact with user metadata in Postgres DB.
- Fine-tuned LLaMA-2 on custom-generated data for recommendations and utilized Redis vector for low latency.

Age and Gender Prediction, Inter-IIT TechMeet, IIT KGP

Dec 2021

- Deployed ESRGANs to detect age and gender from low-quality surveillance video feed.
- Used RetinaFace, and a weighted regression cum classification model for face detection, increasing the F1 score by 0.1.

ACHIEVEMENTS

- Received **Smt. Godavari Devi Award** by IIT Jammu for the highest CGPA in Electrical Engineering.
- **2nd place** in Flipkart Grid Robotics Challenge (Top 0.1% of 9,000 teams nationally)
- **2nd prize** in SEED Pitch Competition among 20 regional teams in Jammu and Kashmir.
- Ranked **first** in Inter-IIT Tech Meet 10.0-Bosch Age and Gender Detection out of the 23 top IITs
- **Semi-finalist** in Google Girls Hackathon 2023 (Top 2.5% nationwide).
- **Top 5** among 75 teams pan India in the SERB-INAE Hackathon organized at Jadavpur University.

LEADERSHIP

Student Head Representative, Career Development Services, IIT Jammu

Oct 2023-June,2024

- Led a team of 70+, sourced 350+ job opportunities, organized 30+ workshops, launched a CEO Leadership Talk series.
- Automated lead tracking, mass mailing list updates, and report generation, reducing manual processing time by 50% .

Dance Club Coordinator, IIT Jammu

July 2021 - Feb 2022

- Managed a 50+ member team, organized 11+ events, and increased outreach by 80% through collaborations.
- Managed social media promotion, and took funding initiatives to increase engagement and support event execution.